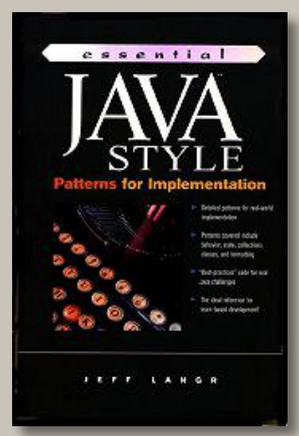
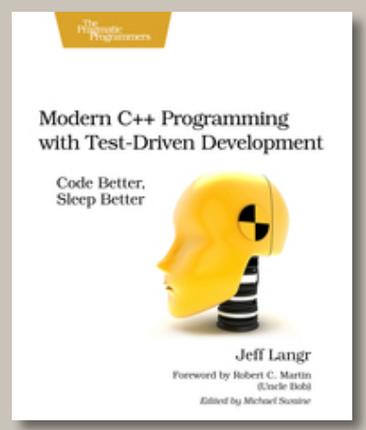
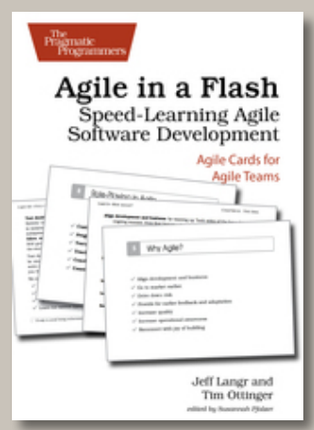
From Mob to Symphony

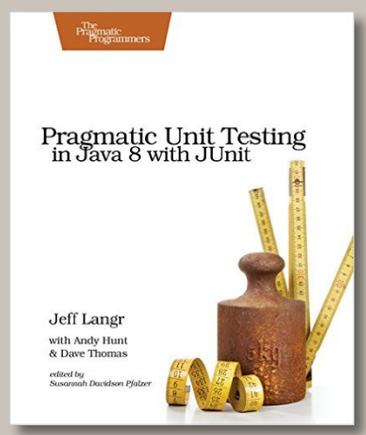
A Mob Composing Workshop

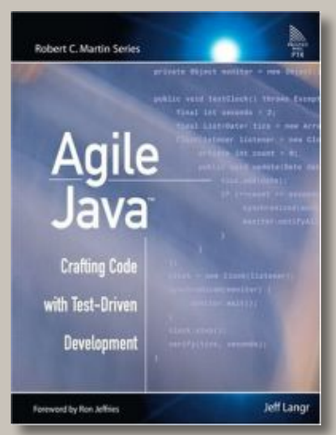
jeff@langrsoft.com
@JLangr

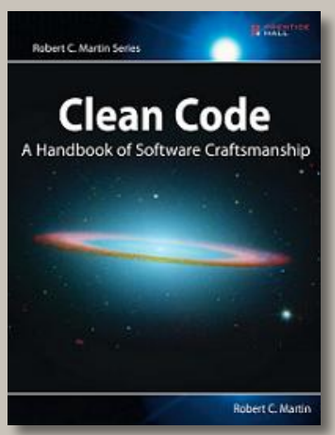












Workshop Overview

- Introductions
- Collaboration and software development
- Mobbing fundamentals
- Challenge overview
- Iterations 1 -> n
  - 20 minute break (~2:40)
- Wrap-up

Law of Two Feet

Intros

- Designate a typist
- For each person, capture:
  - - Name
  - - Where from (company, city)
  - - Primary role
- From the group, capture:
  - - Top three (3) concerns about mobbing

- The typist cannot provide input!

We are uncovering **better ways of developing software**

by doing it and helping others do it.

Through this work we have come to value:

Individuals and interactions

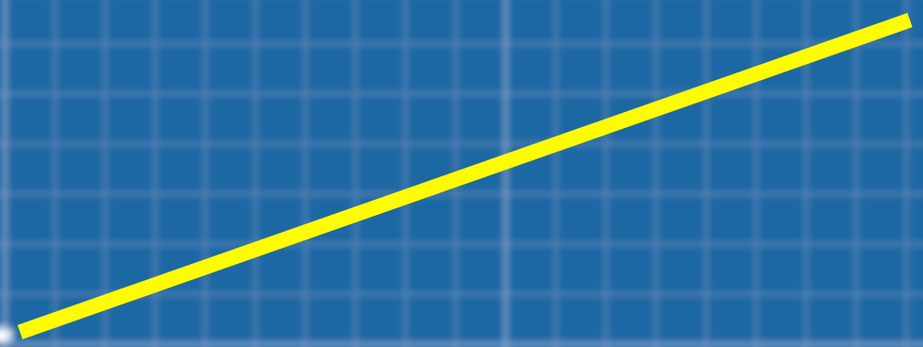
over

processes and tools

The **most efficient and effective** method of conveying information to & within a development team:

Face-to-face conversation

Not this...



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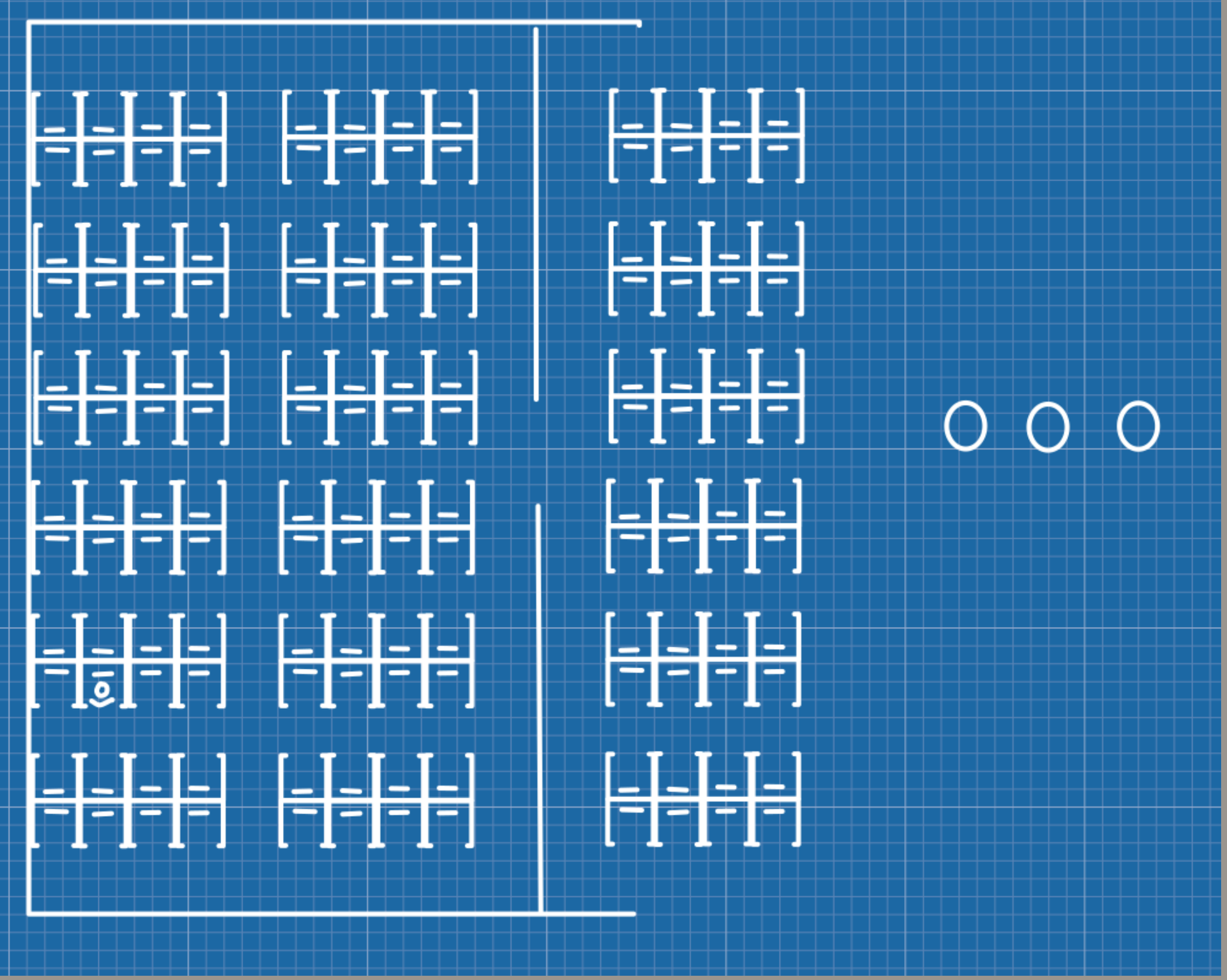

"CUBE FARM"

...or this...

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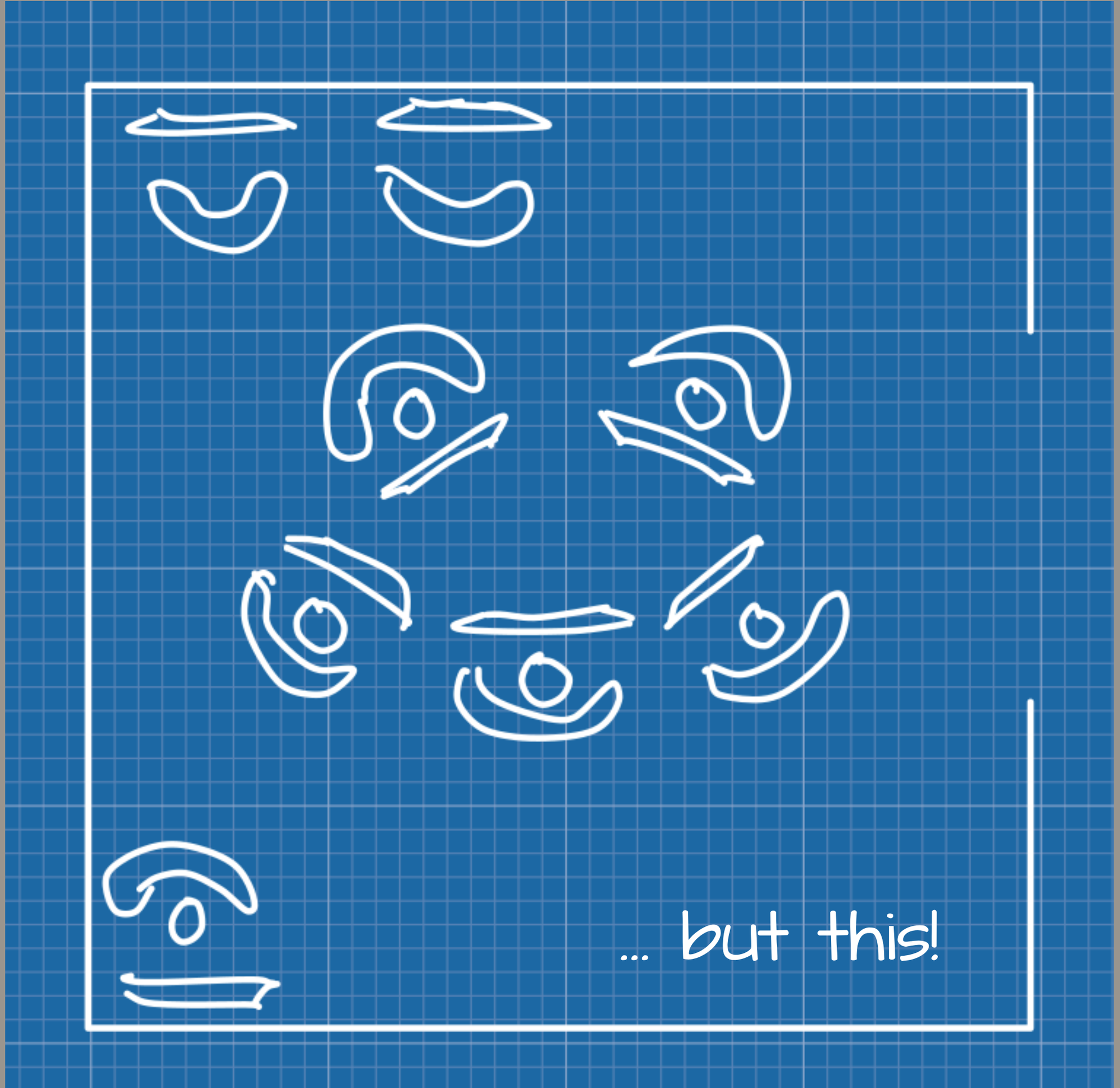

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"Open Plan" Workspace

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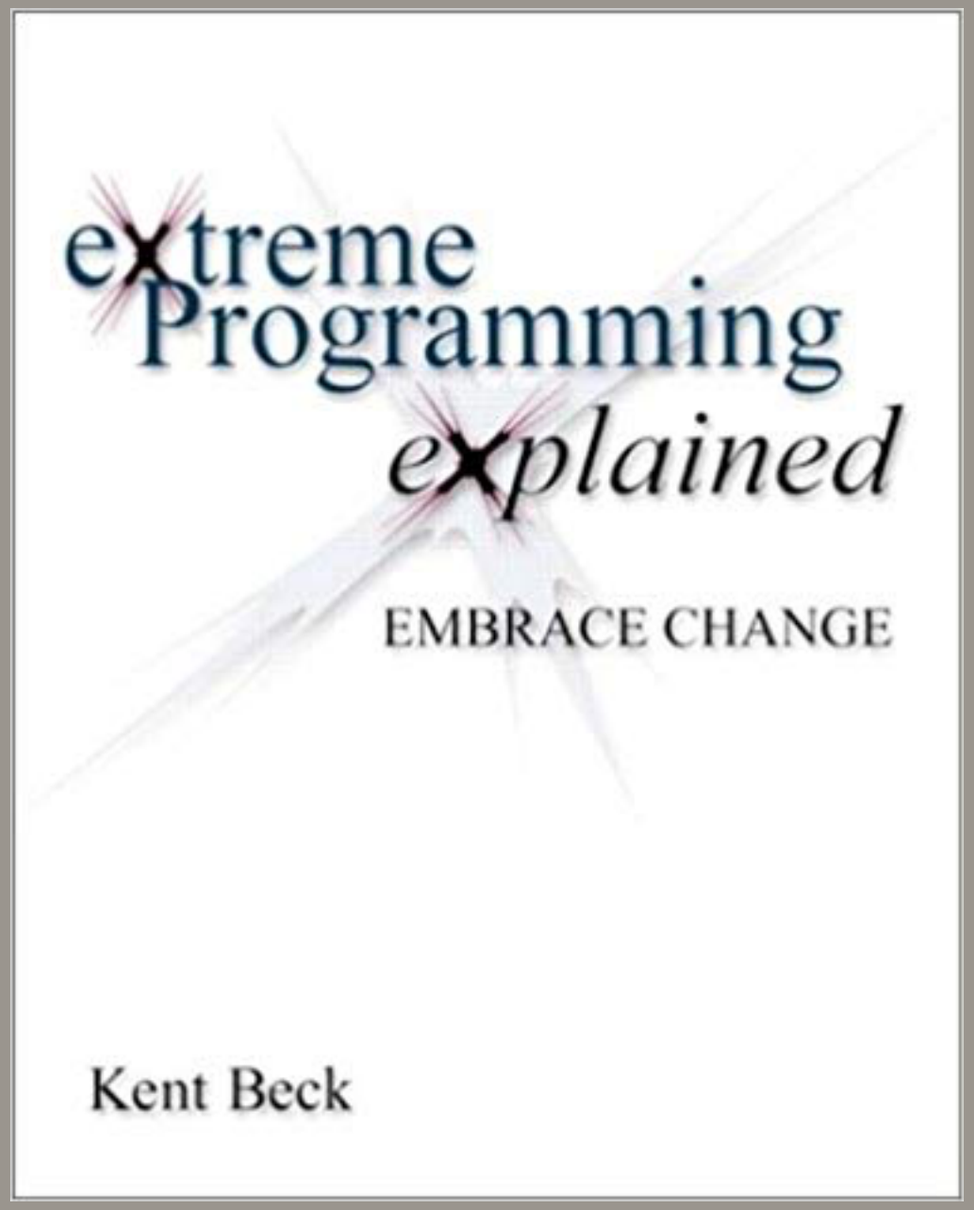
TRANSFORMING IDEA

1999



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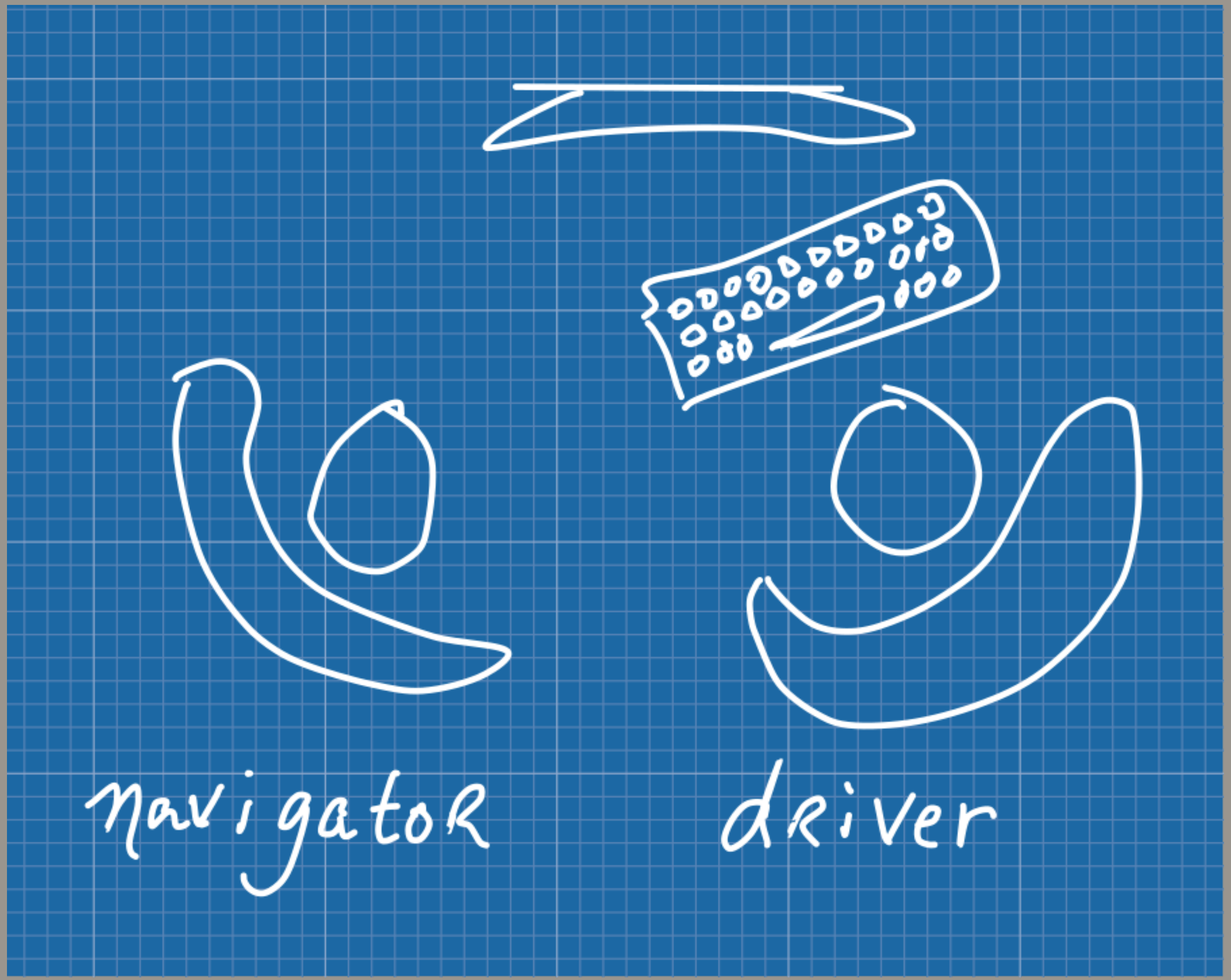
PAIR PROGRAMMING



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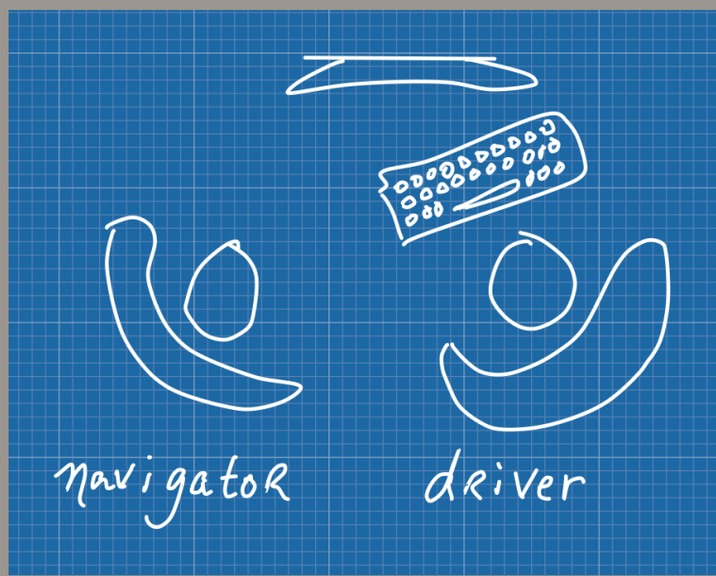
Why?



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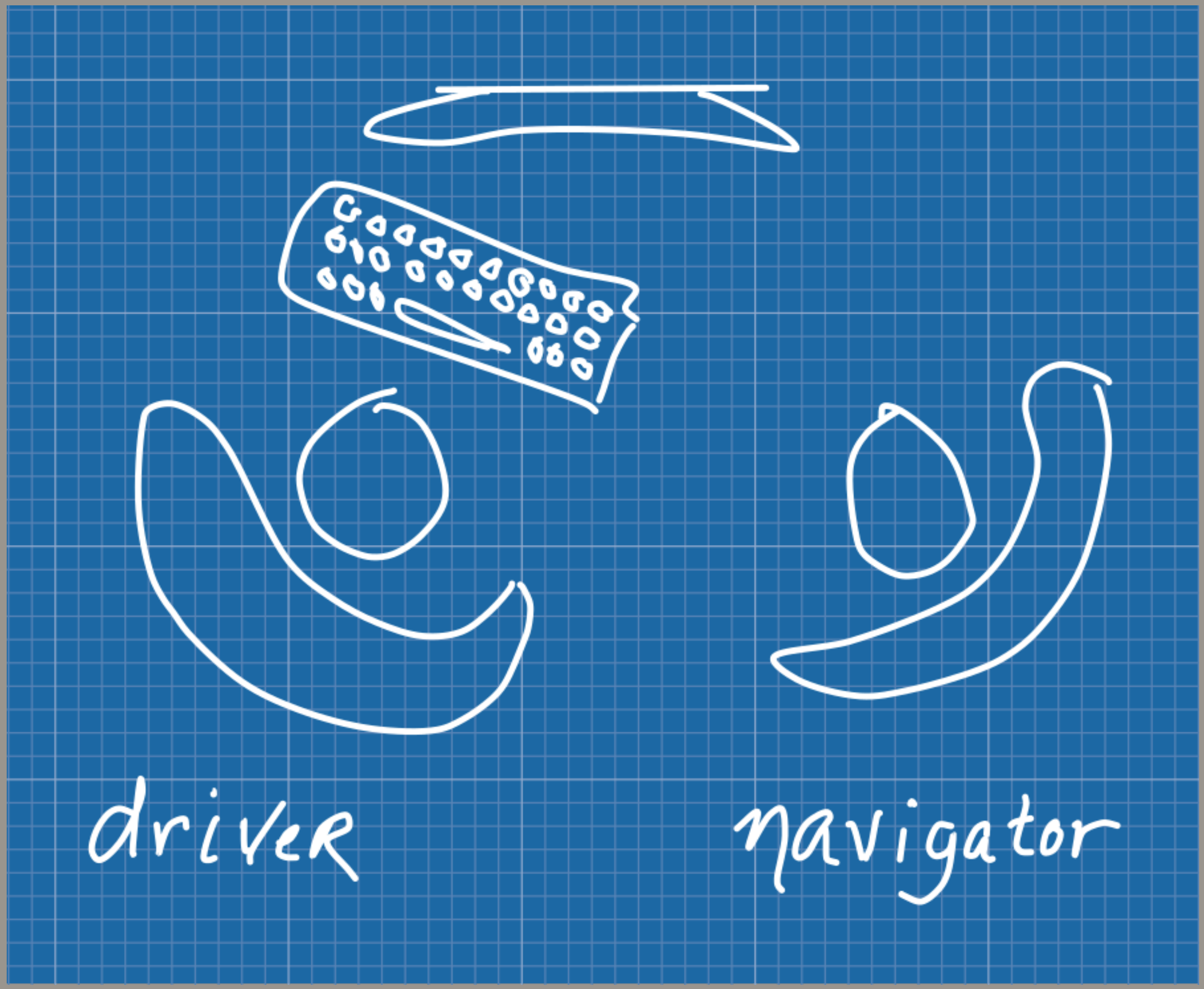
SWITCHING ROLES



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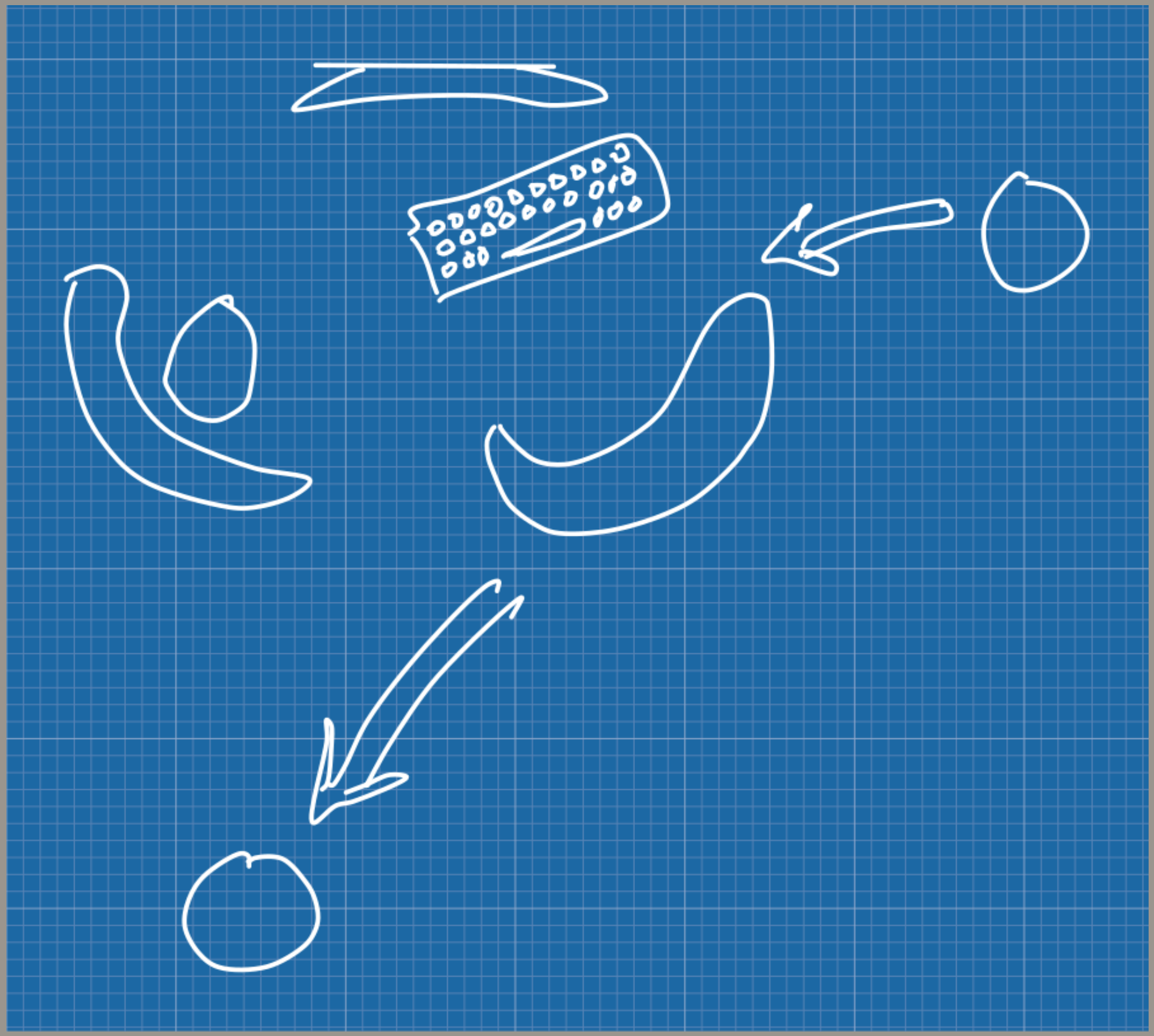

PAIR SWAPPING

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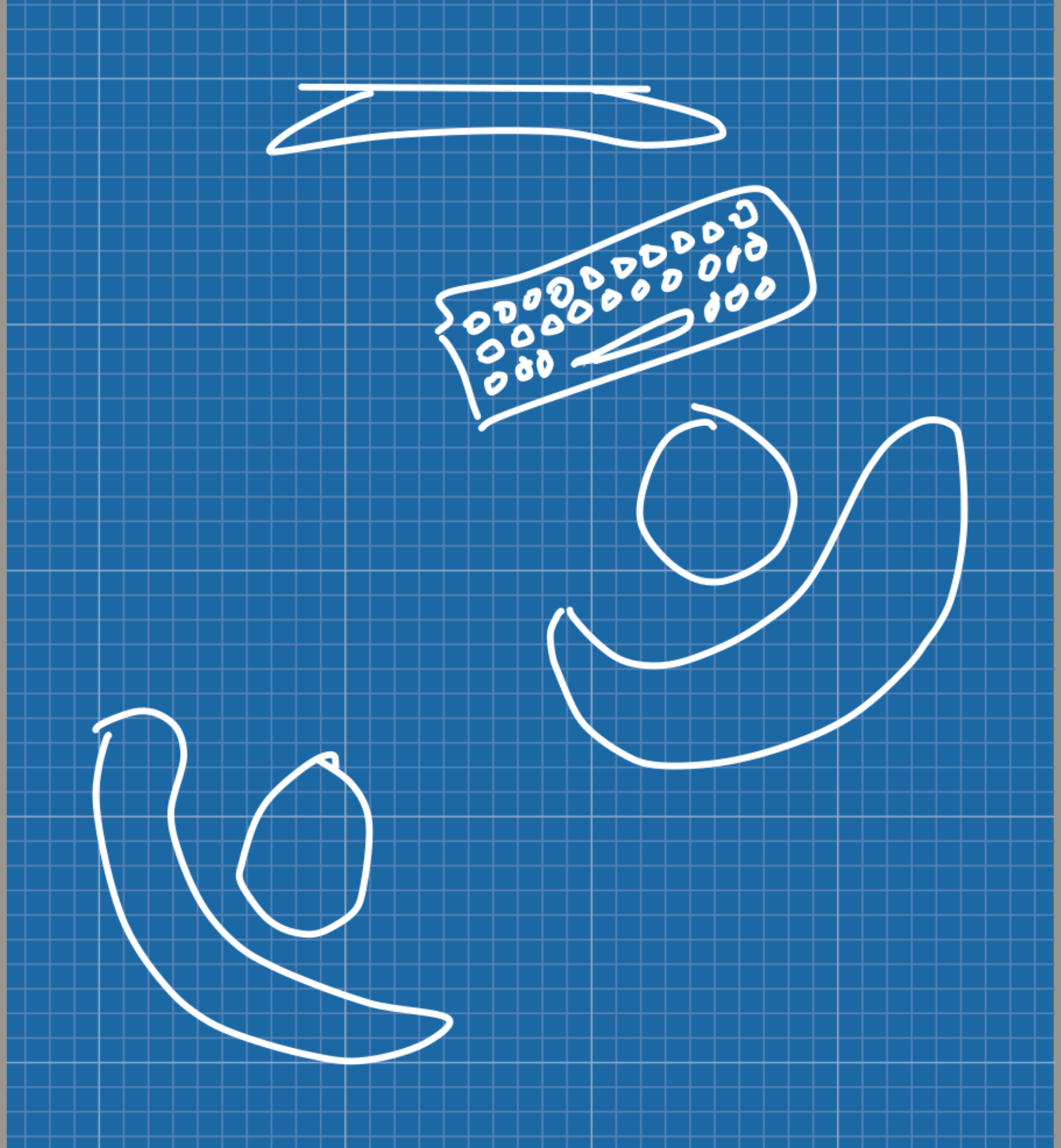
NOT PAIRING!



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SOME PAIRING CHALLENGES

- Fear
- Worker-Rester
- Domination
- Novice-novice
- Hygiene / health
- Interactions
- Physical accommodations
- Accessibility needs
- Shared time
- Individual resistance
- Preferred environments

MOB PROGRAMMING

"All the brilliant people working on the same thing,
at the same time,
in the same place,
and on the same computer."

Photo courtesy Agile Ailliance.

-- Woody Zuill

http://mobprogramming.org

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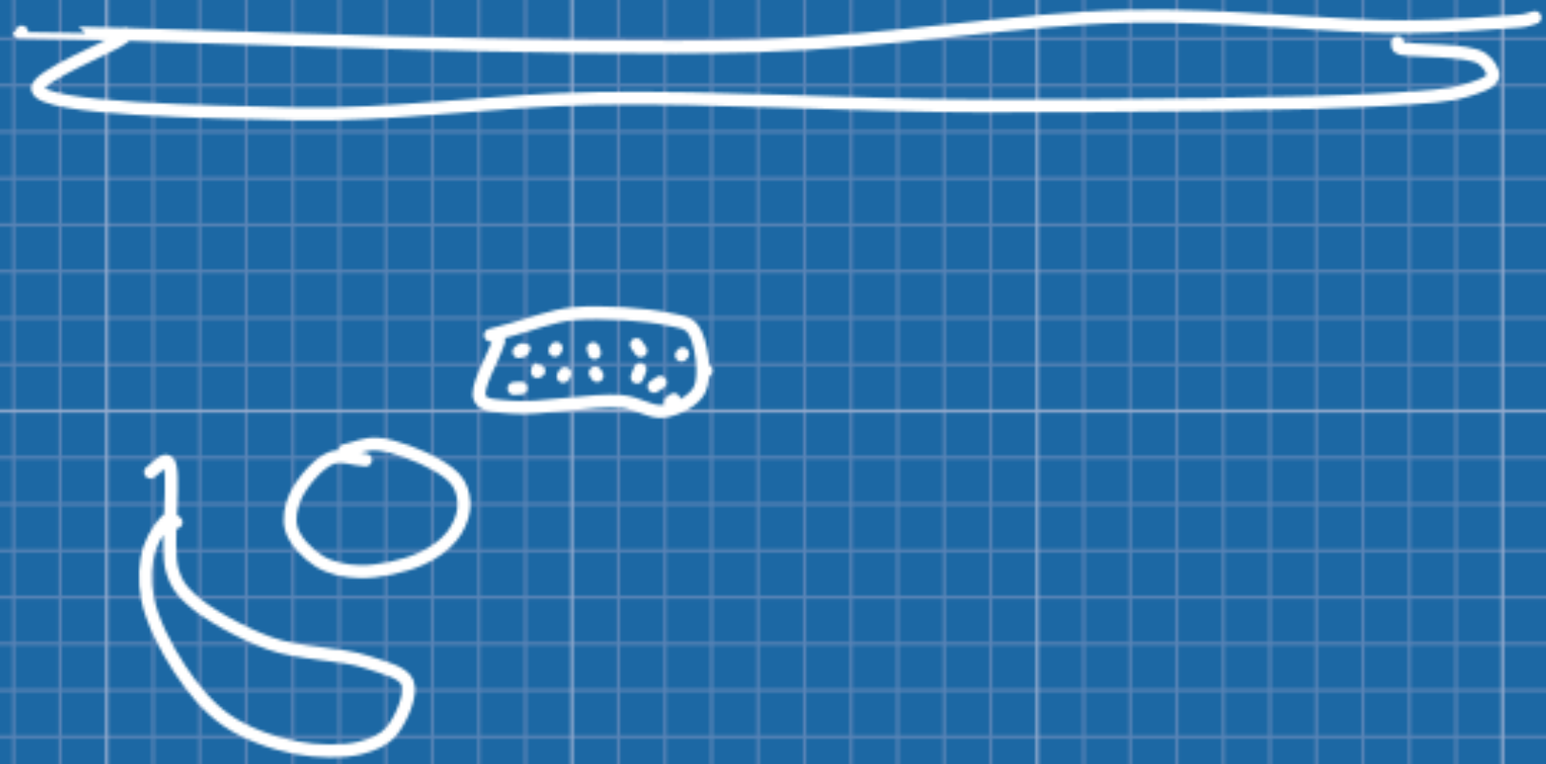
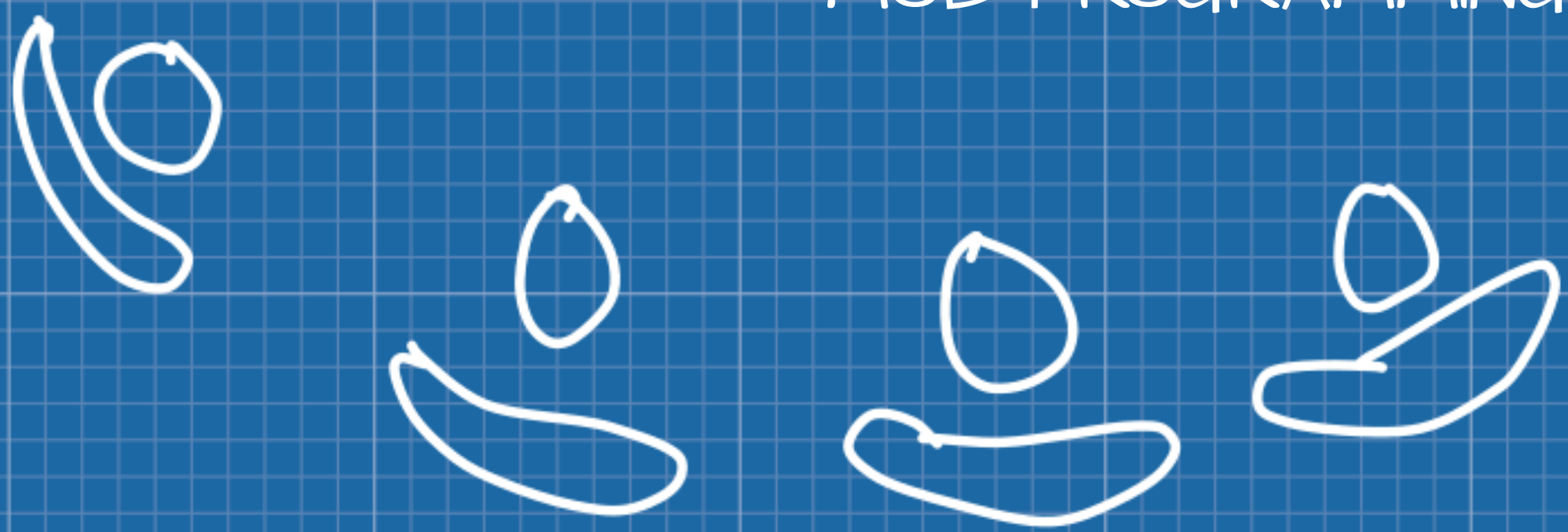


MOB PROGRAMMING



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Business people and developers must work
together daily throughout the project.
The best architectures, requirements, and designs
emerge from self-organizing teams.

The Right Tool for the Job

2 Rules

+

Comfort

+

Ongoing Attention

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Rule #1: DRIVER DOESN'T NAVIGATE

"For an idea to go from your head into the computer it MUST go through someone else's hands."

-- Llewellyn Falco

Strong Style Pairing

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http://llewellynfalco.blogspot.com/2014/06/llewellyns-strong-style-pairing.html

Rule #2: TIMED ROTATION

Mobster

http://mobster.cc

Howler Timer (MacOS)

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"FEAR," Courtesy Kevin B 3, https://www.flickr.com/photos/22850192@N03/8290599649/
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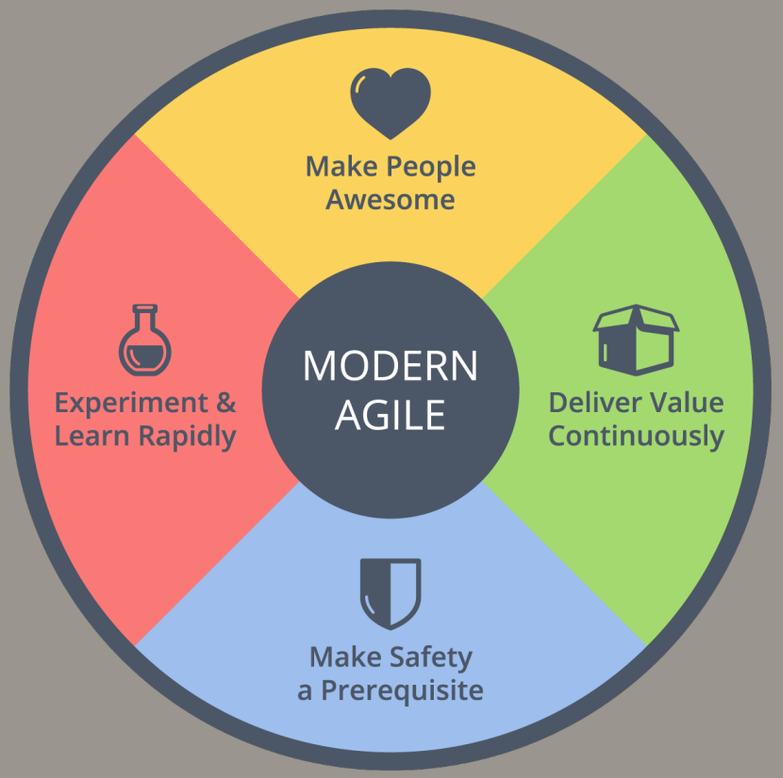
Comfort & Safety

Breaks

Comfortable environment

Safety

?



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Courtesy Industrial Logic, http://modernagile.org

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Ongoing Attention

Mob fatigue

New folks

Interviewing

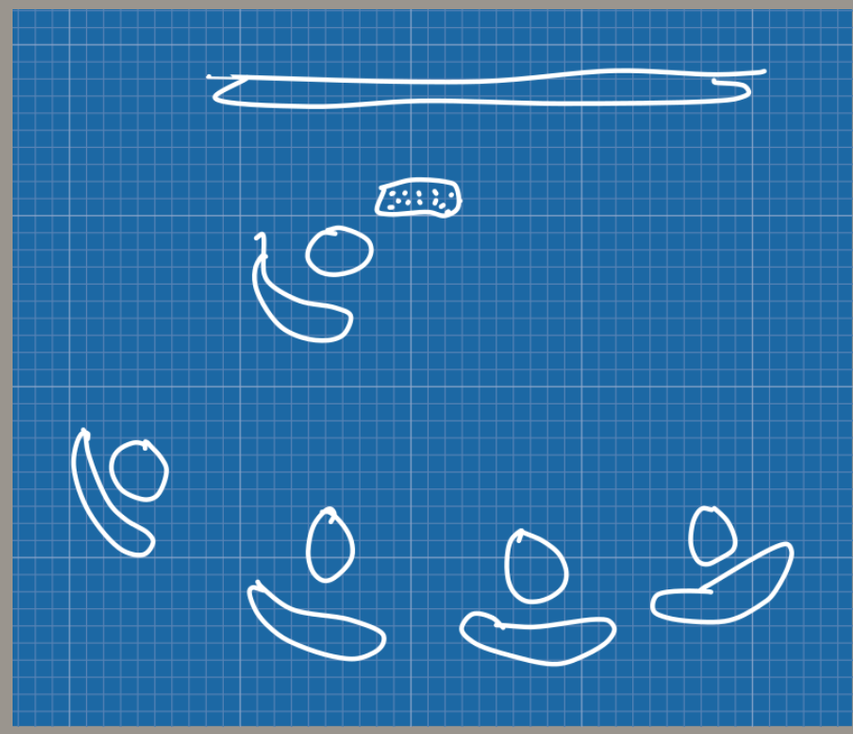
When / when not

Retrospecting

Interruptions

"We go faster."

Why?



Or why not.

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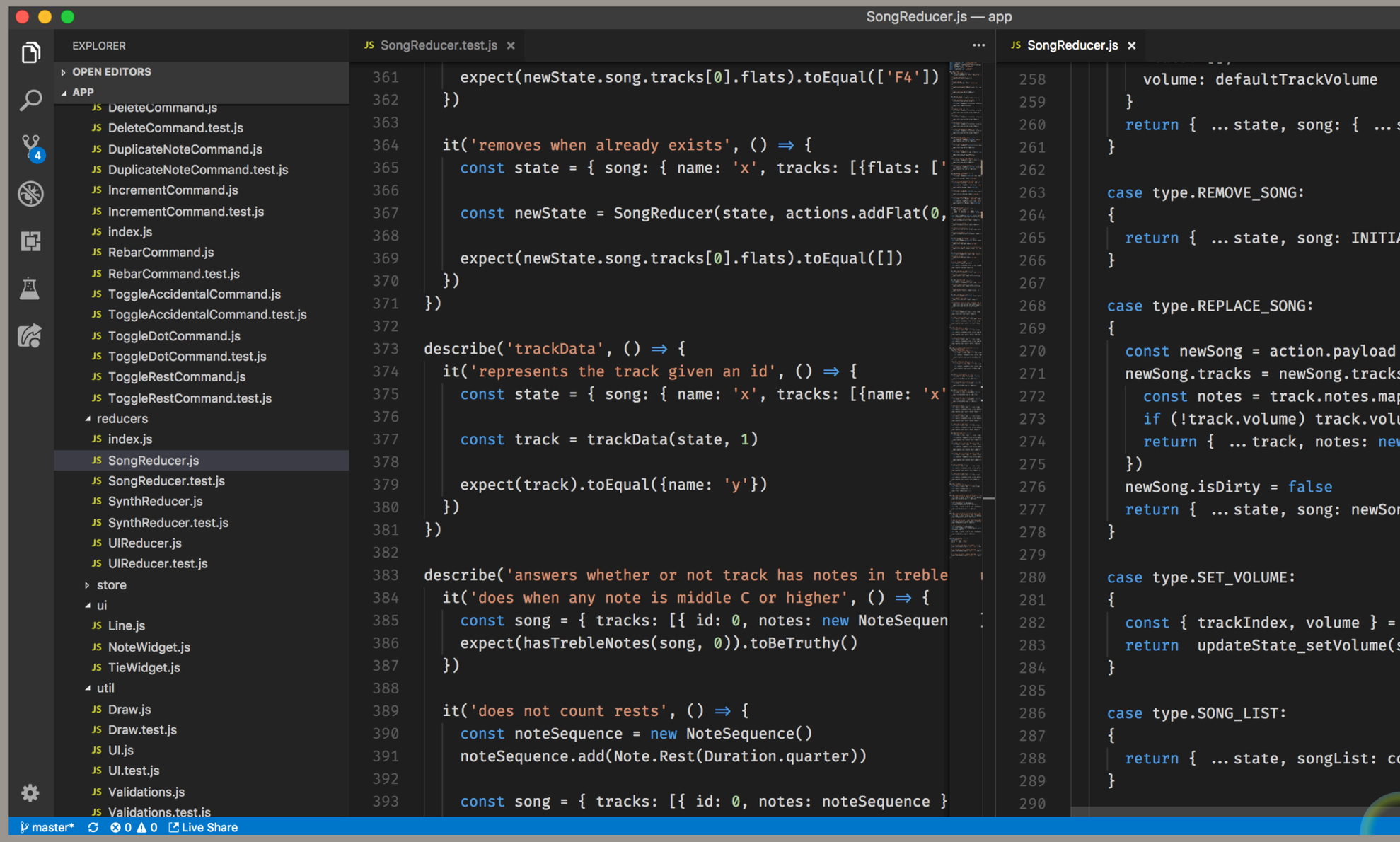
The Workshop

What's like software development?

Q.



Key characteristics of software development?

Key challenges in software development?

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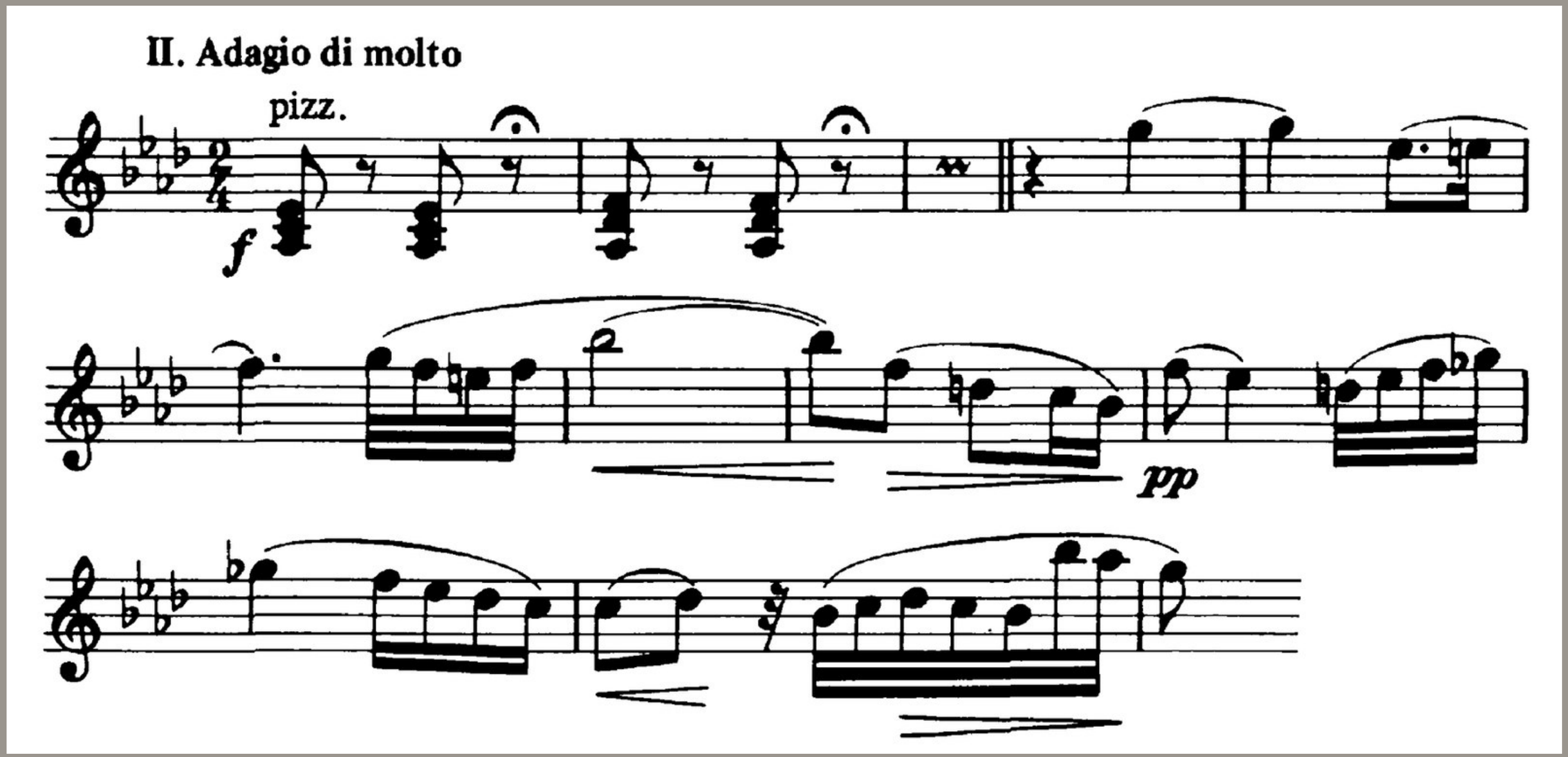
How is composing *similar to* software development?



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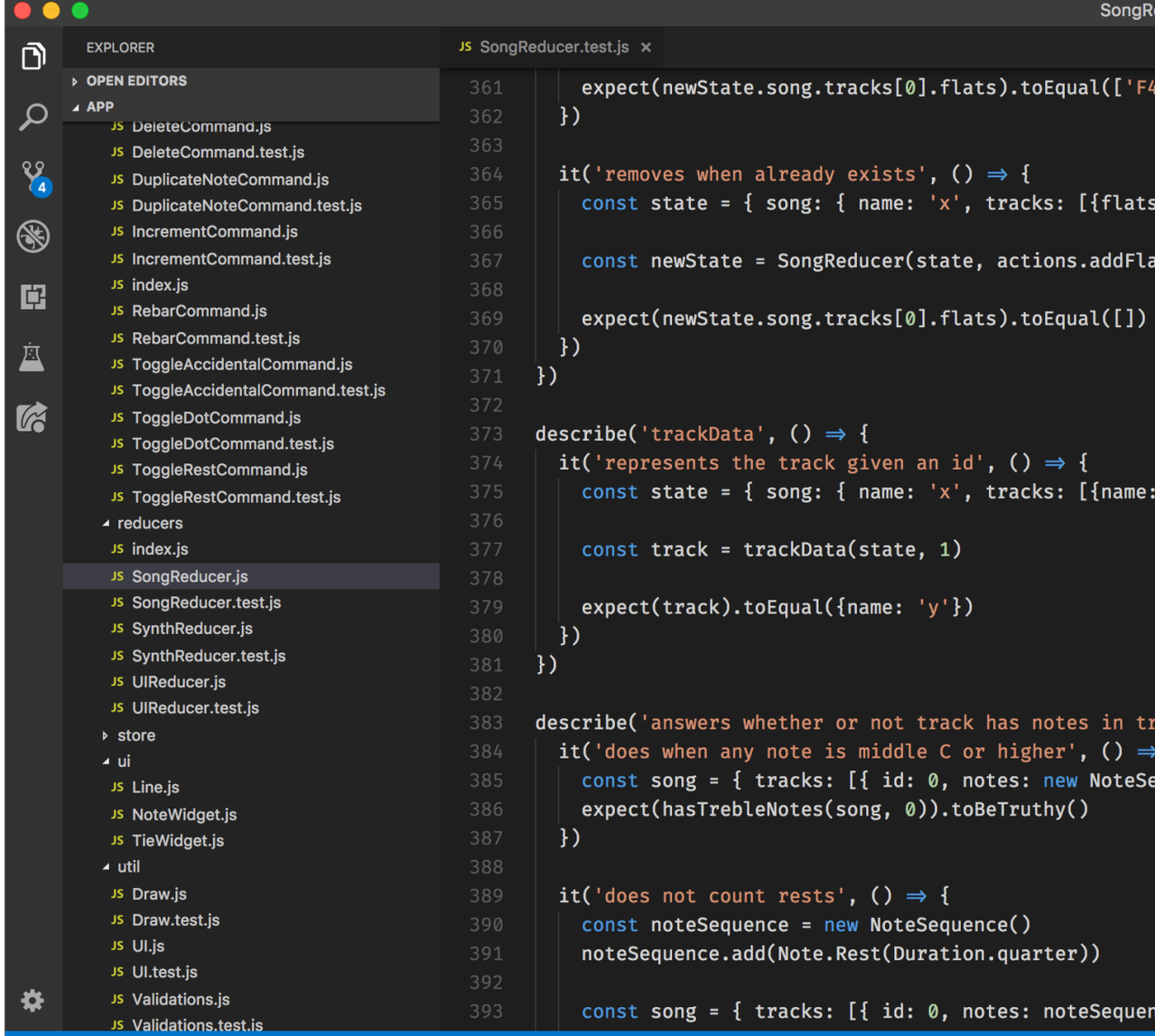

How is it *different from* software development?

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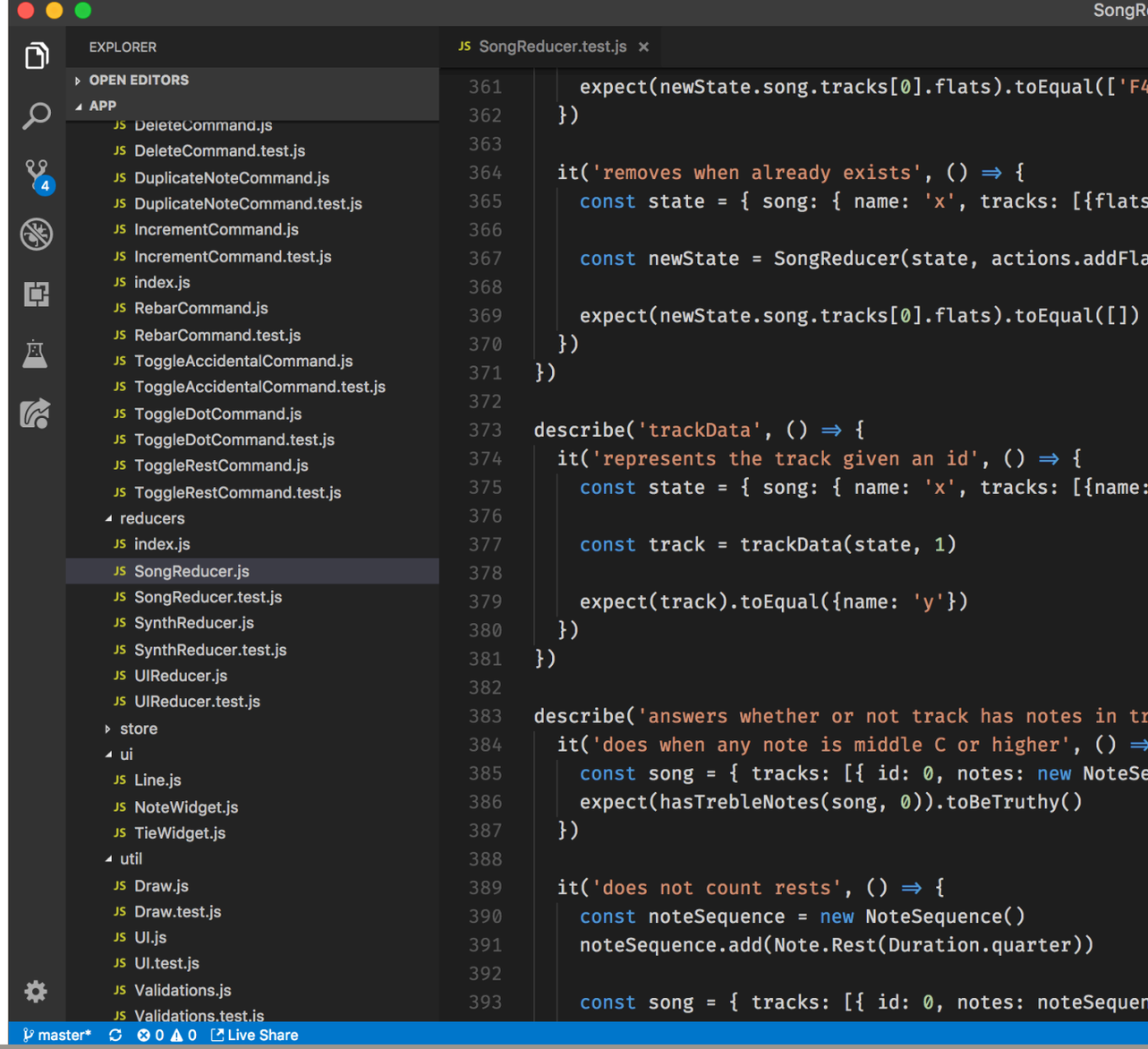

Team or individual activities?

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What Do We Know About Music?

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https://www.pexels.com/photo/people-coffee-meeting-team-7096/

Translation Levels

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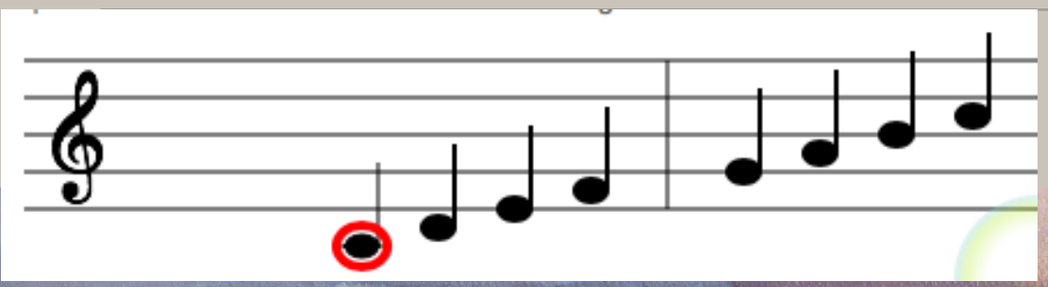
An Iteration

- Overview of next mobbing objective / rule
  - What to look for / think about
- Overview of task
- ~20 minutes of development time
- Demos
- Review of objective
- Mini-retro

Tasks

- Installing Geektone
- Exploring Geektone (twinkle twinkle)
- Melody
- Chorus
- Tension / release
- Rhythm and "space"
- Counter-melody
- Intro
- Outro

Geektone & Music Basics



Notes

- Duration

1/4 quarter

1/2 half

1 whole

1/8 eighth

1/16 sixteenth

- Dotted

One-and-a-half times normal duration

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Rests

- Duration

1/4	1/2	1	1/8	1/16
quarter	half	whole	eighth	sixteenth

Notes Have Pitch

staff

C D E F G A B C

do re mi fa sol la ti do

a scale

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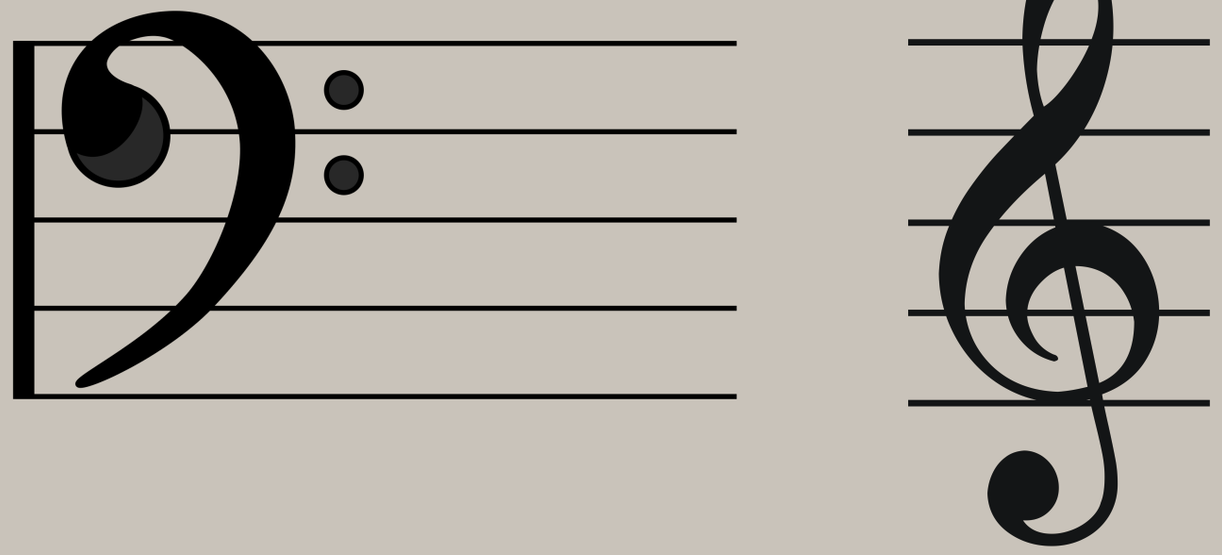

Bass & Treble Clefs

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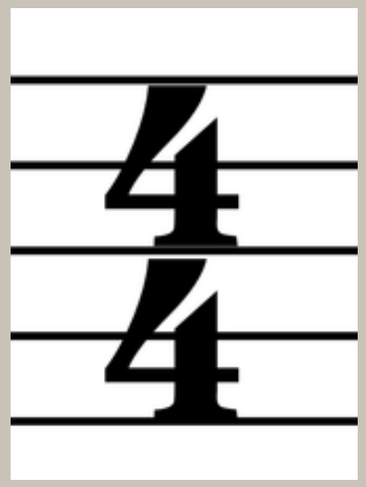
Sharps & Flats

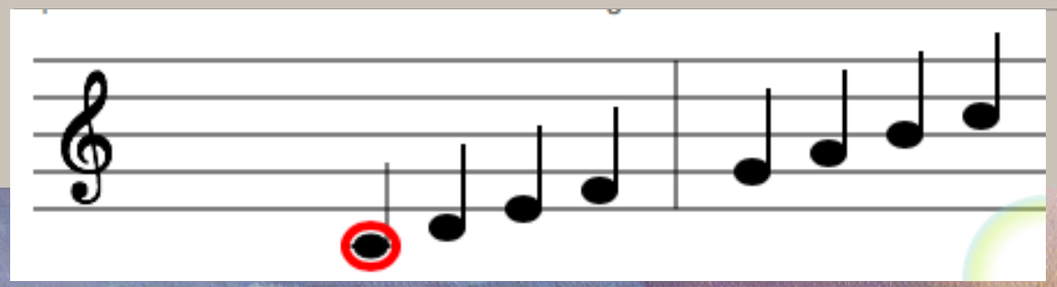
- Sharp: half step up (semitone)
- Flat: half step down
- Applied to all notes of same pitch!

Measures & Beats



- 4/4 time:
  - 4 beats per measure
  - quarter note gets the beat

Composing Challenge



The Challenges

- 45 or 60 seconds
- Television commercial
  - Feel-good product ad, emotive charity ad, medical malady ad, …
- Movie title track
  - Horror, comedy, drama, adventure, …
- Television montage
  - Uplifting; increasingly desperate; tear-jerking, …

- Choose challenge (5 min)
- Discuss theme (5 min)
  - Agree on the core emotion(s) to provoke
    - "anxiety," "joy," "relaxation"
  - Agree on a few of goals
    - "punchy," "meandering," "increasingly frantic"

Go !

Review.

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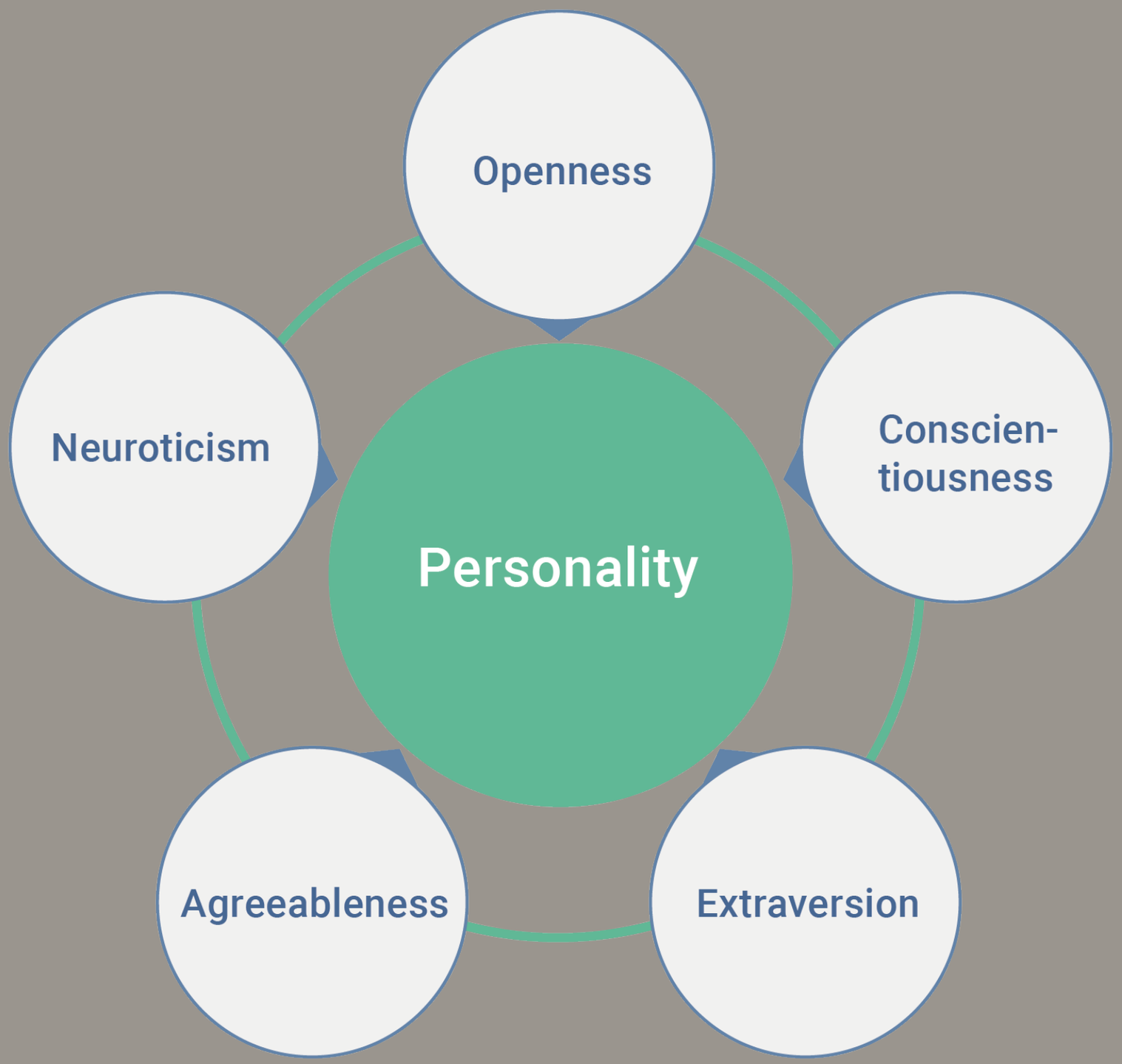
Reflect.



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https://en.wikipedia.org/wiki/Big_Five_personality_traits

"Pomodoro Kitchen Timer for Activity Logging," courtesy Andy Roberts
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Pomodoros

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https://en.wikipedia.org/wiki/Pomodoro_Technique

Song Structure

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Basic Melody

(A) common thread through a song

- Range: An octave and a half
- Perhaps contains some repeating elements
- Mostly "stepwise" with occasional leaps
- Climactic moment
  - Highest notes?

Verse

- Tells a story -- often new lyrics
- Details of the song
- Supports the chorus

Intro	Verse	Chorus	Verse	Chorus	Bridge	Verse	Chorus	Outro

Chorus

- Repeated verse / phrase: a motif

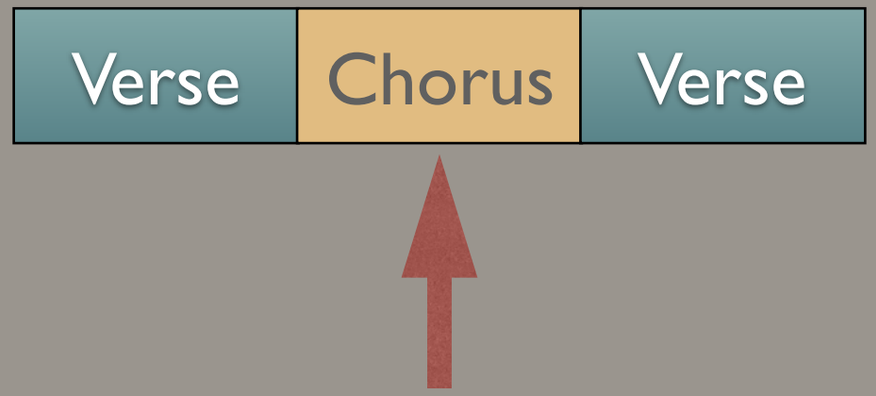
Intro	Verse	Chorus	Verse	Chorus	Bridge	Verse	Chorus	Outro

- Short and simple!
- Repetition + simple variants
- Punctuate
  - Volume
  - Silence
  - Add'l instruments

Pomodoro: Compose a basic song

~10-15 seconds

~10-15 seconds



shorter

Rules Reminder

Five minute switch

Keyboardist drives *only*

Mob !

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https://www.pexels.com/photo/chairs-device-electronics-eyeglasses-1181376/

Review.

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Reflect.

Driver Tip

Overwhelmed? Feeling micromanaged?

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https://www.pexels.com/photo/bow-tie-businessman-fashion-man-1702/

Mob Tip

"This is a good place for an example." -- Brian Marick

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Courtesy YogeshMishra Maharaja, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=64491074

Flow

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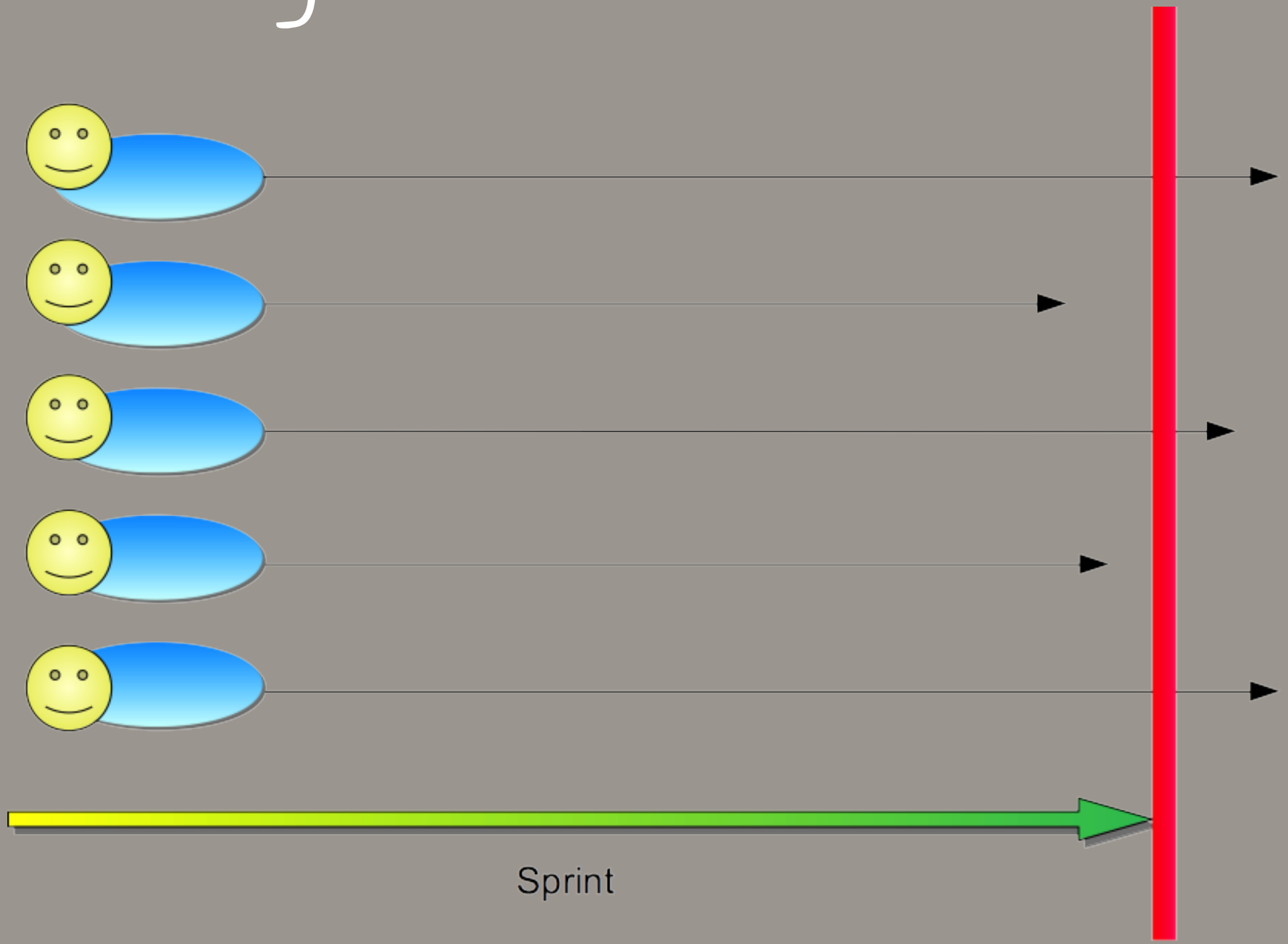
https://www.pexels.com/photo/architecture-auto-automobiles-bridge-210182/

Classic Agile Iteration



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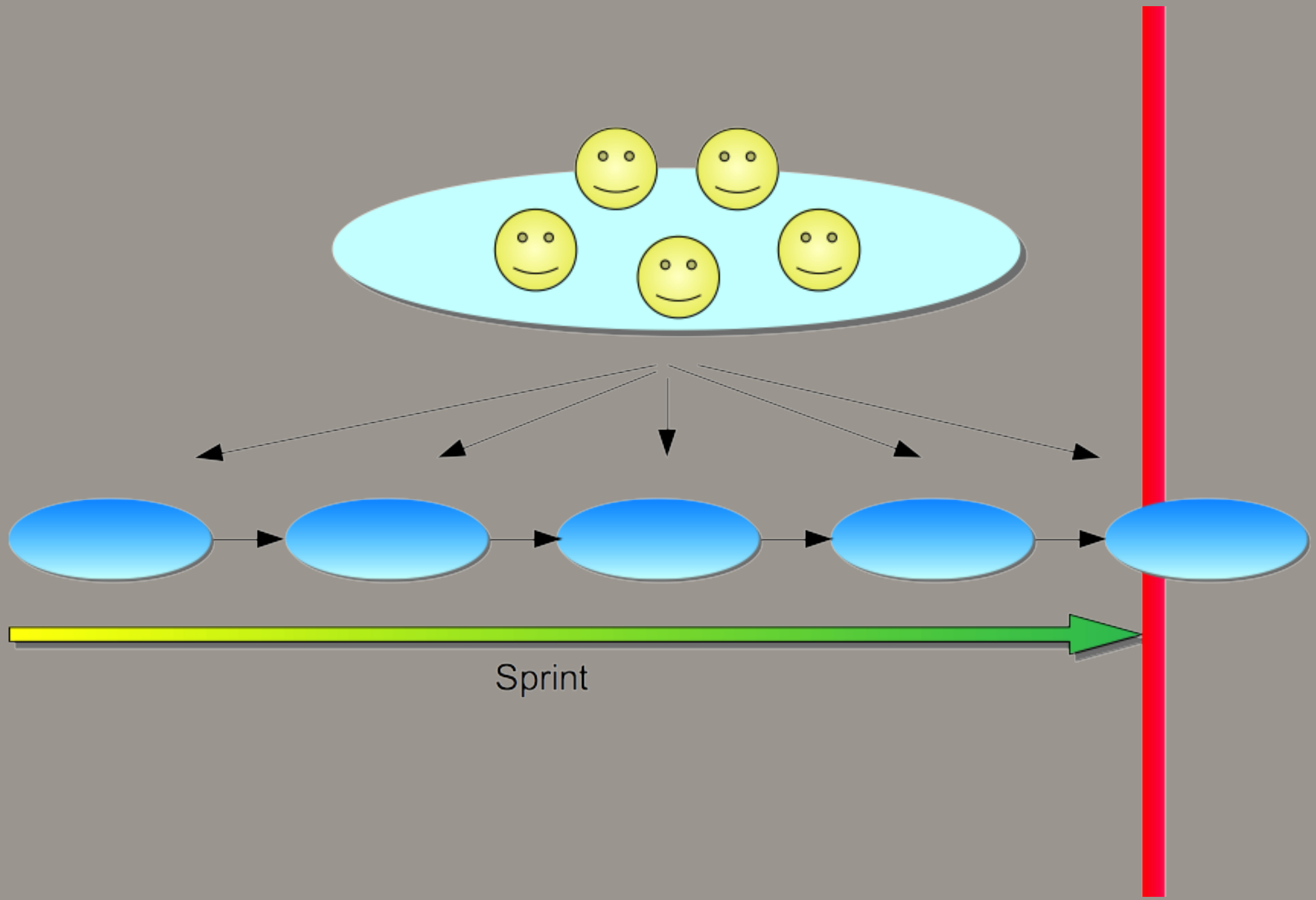
Reduced WIP



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Creating "Space"

- Silence is golden
- Rests can:
  - add interest to melodies
  - create interesting rhythms
  - create suspense

Rhythm

- Brings character to the music
- Keeps the song together
- … and propels it
- Makes us want to move

- Listen to some basic rhythms
- Start simple: be repetitive
- Tweak it!
- Clap it out

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Pomodoro: Rhythm & Space

- Some iterate, some increment
- Seek to introduce space in the verse / chorus
- Create a new track for rhythm

Rules Reminder

Five minute switch

Driver does not navigate

Mob !

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https://www.pexels.com/photo/chairs-device-electronics-eyeglasses-1181376/

Review.

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Reflect.

Driver Tip

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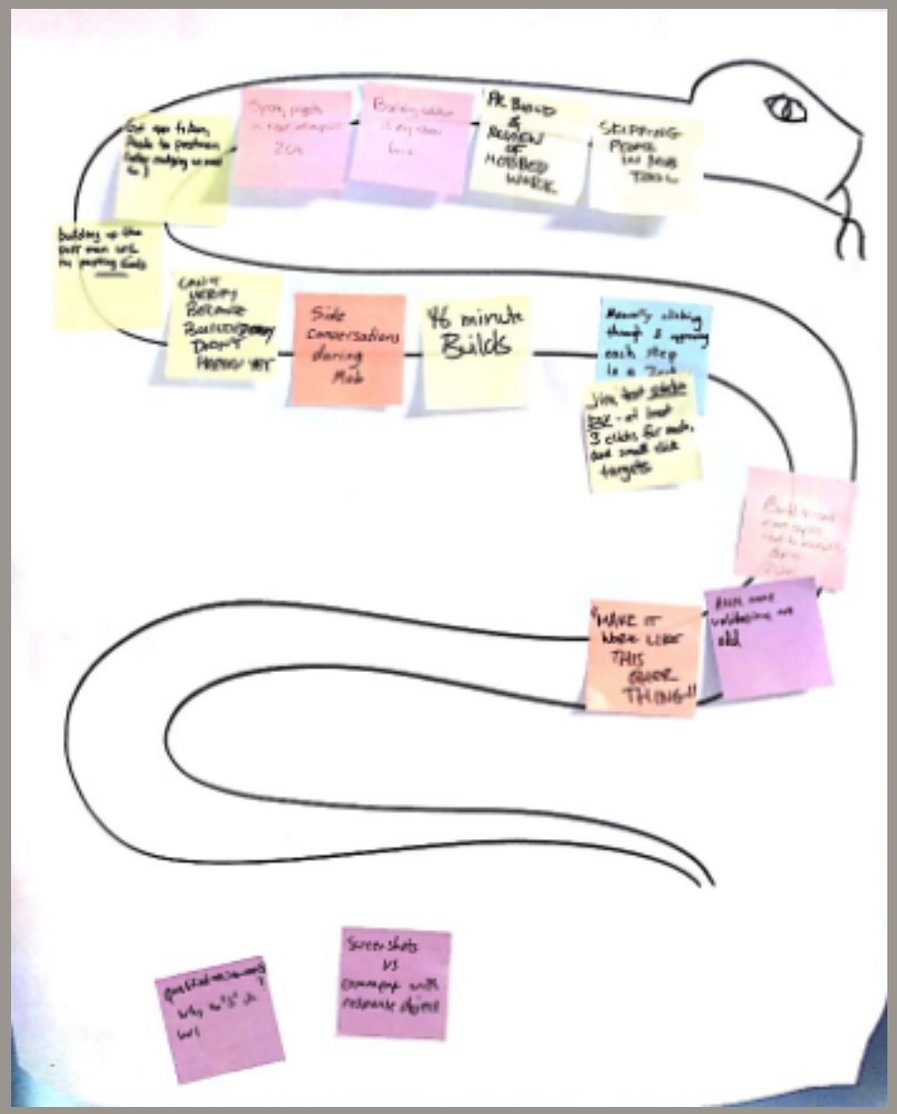
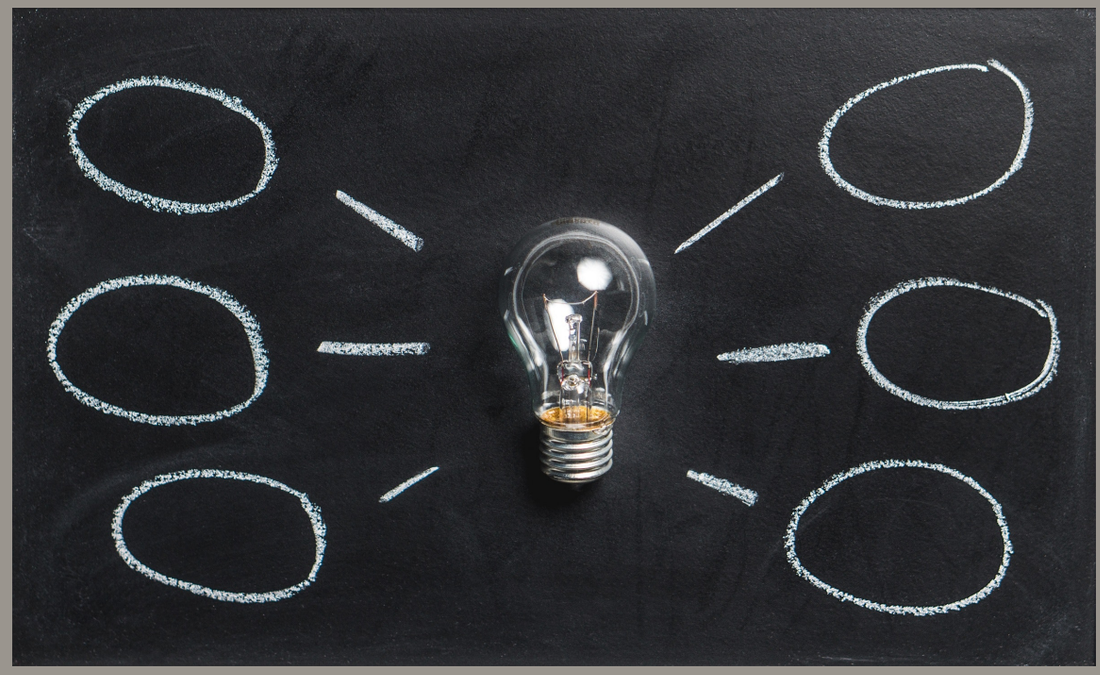
https://www.pexels.com/photo/parking-lot-753865/

Mob Tip



Waste Snake

Courtesy Industrial Logic / Tim Ottinger
https://www.industriallogic.com/blog/a-few-tips-for-mob-programming/



Lessons Learned

https://www.pexels.com/photo/abstract-blackboard-bulb-chalk-355948/

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https://www.pexels.com/photo/bow-tie-businessman-fashion-man-1702/

Tension / release

- Building unrest / conflict
- Needs release!
- Example: Chord progressions

- Rhythmic tension
- Melodic tension
- Bridge tension
- Instrumental tension
- Dynamic tension

Pomodoro: Introduce tension

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Rules Reminder

Five minute switch

Driver does not navigate

Mob !

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https://www.pexels.com/photo/chairs-device-electronics-eyeglasses-1181376/

Review.

@JLangr

Reflect.

Additional Elements for Pomodoros

Counter-Melody

- Subordinate to main melody
  - Uses same scale
- Sounds good alone
- Should oppose, not overlap

Intro

- Grab the listener's attention
- Establish style / mood

Intro	Verse	Chorus	Verse	Chorus	Bridge	Verse	Chorus	Outro

- Perhaps use materials from rest of song
- Contrast:
  - Stripped down
  - Slower / faster
  - Grabby hook!...

Outro / Conclusion

- Inverse of intro?
- Wrap things up

Intro	Verse	Chorus	Verse	Chorus	Bridge	Verse	Chorus	Outro

- No rules
- Simple, might repeat elements of chorus / verse

Bridge

- Optional transition
- Pause / reflect on earlier parts of song
- Prepare listener for climax

Intro	Verse	Chorus	Verse	Chorus	Bridge	Verse	Chorus	Outro

- Opposing melody, chord progression
- Deepen emotional impact
- Connect smoothly

Wrapping Up

Some Tips

Law of two feet

Established core hours

Regular breaks!

Be there.

Designated navigator if needed.

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https://www.pexels.com/photo/close-up-of-hand-over-white-background-255527/

Why Faster

- Important discussions happen before code is integrated
  - Before it's too late
- Fewer / simpler rituals

  (standups, merging, pull requests, defect tracking, etc)
- WIP of one--focus on delivery
- All the info in the room
- All the wisdom
- Flow
- Everyone understands the code
  - And why / decisions what went into it
- Consistent solutions
- Adherence to quality standards more likely
- Less rework (defects)

Things to Try

Guests

- experts
- managers
- ...

Warm-ups / katas
Speed challenges
Skills learning session

http://www.rtqe.net/ObliqueStrategies/

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"Kata," courtesy https://www.flickr.com/photos/superwebdeveloper/
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"Beautiful and dusty," courtesy https://www.flickr.com/photos/80435089@N08/
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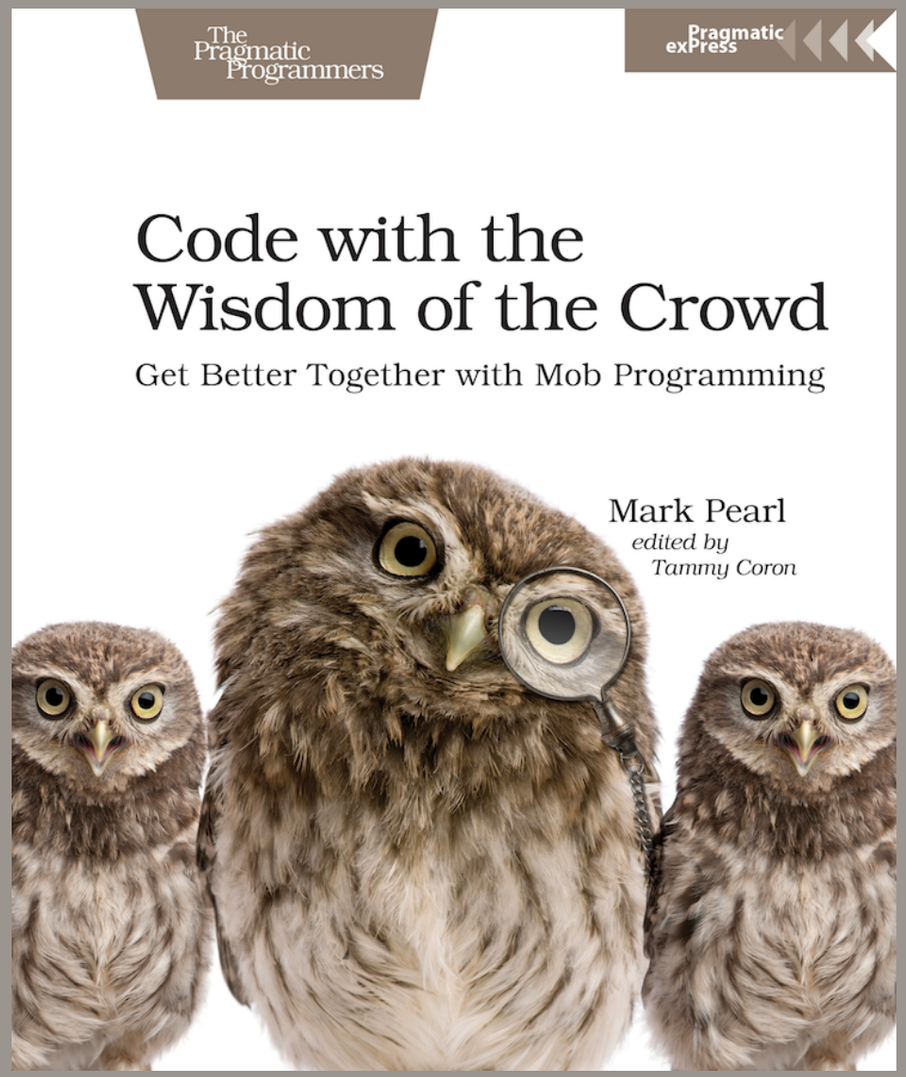
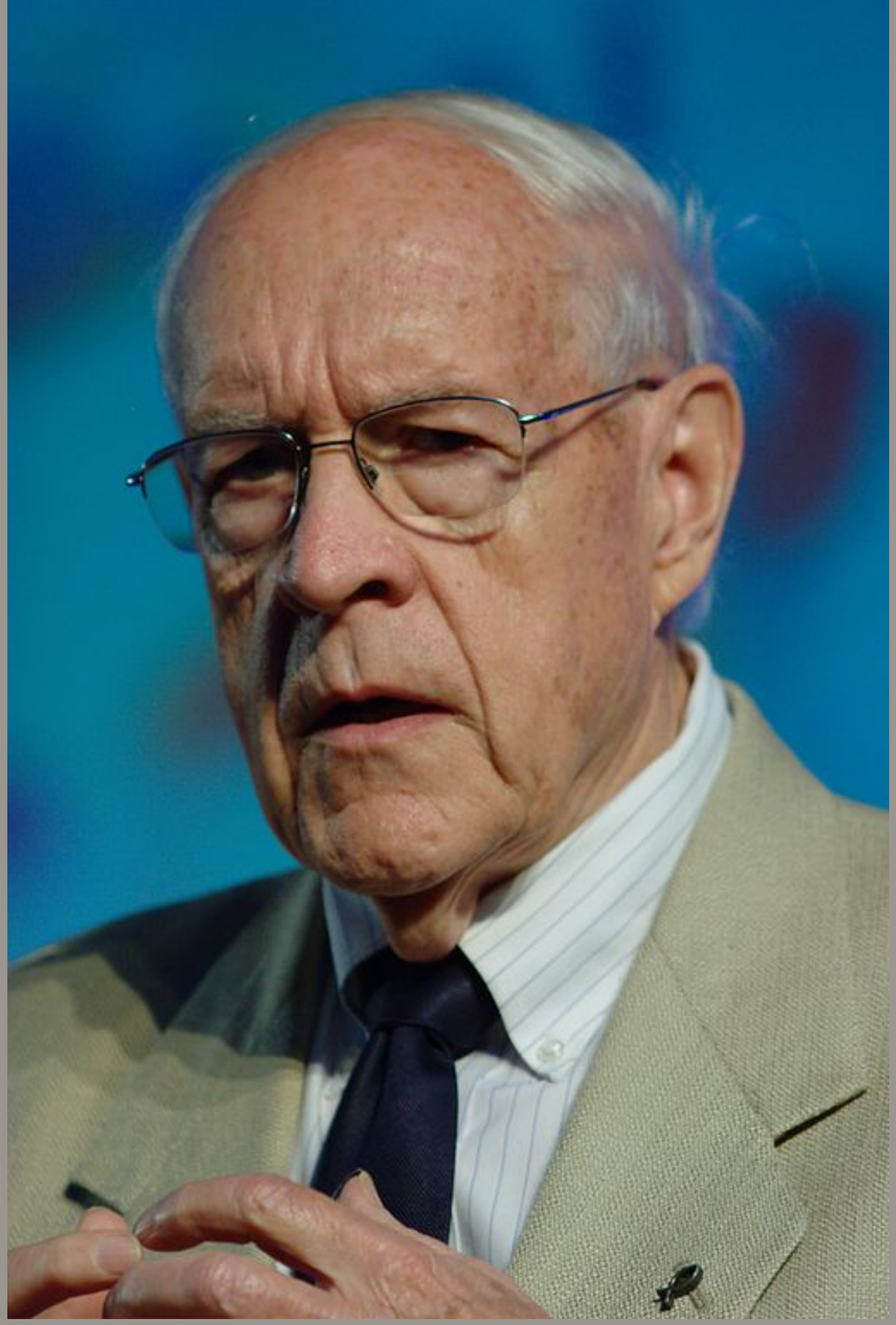
Some Resources

http://mobprogramming.org

https://www.industriallogic.com/blog/a-few-tips-for-mob-programming/

also:





https://commons.wikimedia.org/wiki/File:Frederick_Brooks_IMG_2279.jpg

Brooks' Law

"Adding manpower to a late software project makes it later."

The Mythical Man-Month, 1975

@JLangr

What Next

Could your team do this?

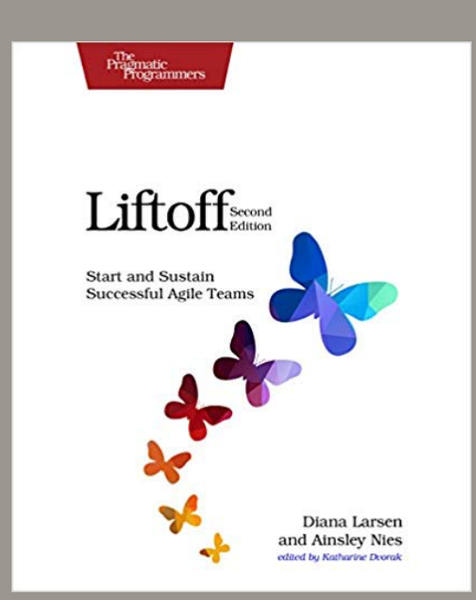
Agree to run experiment

"Mobbing Tuesdays?"

- or M-W, Tu-Th, 9-11:30, ...

Create working agreements

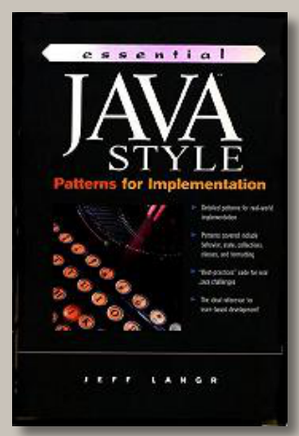
Solicit private feedback

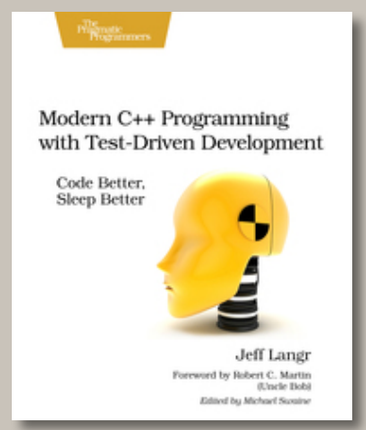


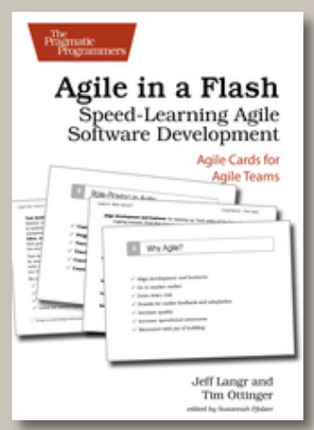
Have fun with it

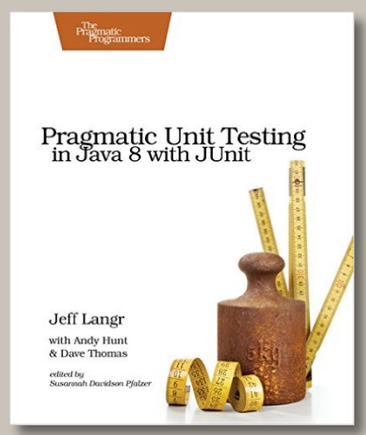
Thank you!

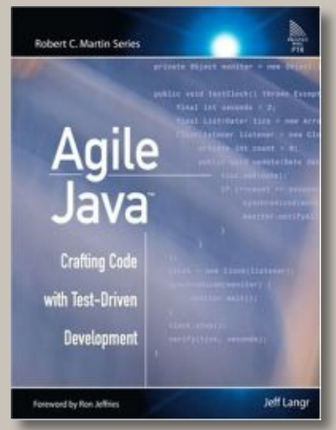
Langr
Software Solutions
SOFTWARE TRAINING/CONSULTING
TDD
AGILE

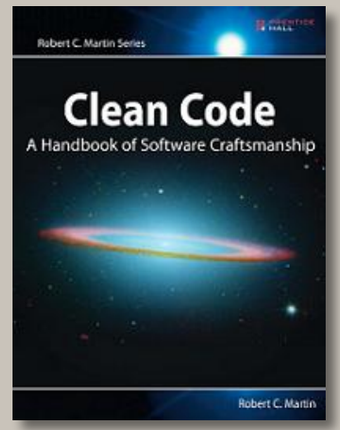

essential
JAVA STYLE
Patterns for Implementation
JEFF LANGR


The Pragmatic Programmers
Modern C++ Programming with Test-Driven Development
Code Better, Sleep Better
Jeff Langr
Foreword by Robert C. Martin (Uncle Bob)
Edited by Michael Swaine


The Pragmatic Programmers
Agile in a Flash
Speed-Learning Agile Software Development
Agile Cards for Agile Teams
Jeff Langr and Tim Ottinger


The Pragmatic Programmers
Pragmatic Unit Testing in Java 8 with JUnit
Jeff Langr
with Andy Hunt & Dave Thomas
edited by Susannah Davidson Pfalzer


Robert C. Martin Series
Agile Java
Crafting Code with Test-Driven Development
Foreword by Ron Jeffries
Jeff Langr


Robert C. Martin Series
Clean Code
A Handbook of Software Craftsmanship
Robert C. Martin